

LTAP IN DATABRICKS

THE ARCHITECTURE

POWERING MODERN DATA & AI



**Lake Transactional/
Analytical Processing**

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WHAT IS LTAP - LAKEHOUSE TRANSACTION PROCESSING

LTAP is Databricks' approach to running:

- ◆ Transactional Workloads (OLTP-like)
 - ◆ Analytical Workloads (OLAP)
 - ◆ Streaming Applications
 - ◆ AI & Machine Learning
- on a single Lakehouse platform.

Traditionally, organizations maintained separate systems for transactions and analytics. LTAP eliminates this separation by enabling applications, analytics, and AI to work directly on the same data.

- One copy of data
- Real-time insights
- Lower operational costs
- Faster AI development



THE PROBLEM WITH TRADITIONAL DATA ARCHITECTURES

Most organizations follow this architecture:

Operational Database



ETL Pipeline



Data Warehouse



Data Mart



Machine Learning Platform

Business Impact:

- Slower decision making
- Increased engineering effort
- Higher cloud expenses
- Reduced agility

Challenges

Multiple Copies of Data

Complete ETL Pipelines

Data Synchronization Issues

High Infrastructure Costs

Delayed business insights

Governance Challenges

THE LAKEHOUSE FOUNDATION

LTAP Starts with the Lakehouse

Databricks introduced the Lakehouse architecture to combine the best features of Data Lakes and Data Warehouses

The Lakehouse provides a single platform where data engineering, analytics, BI, data science, and AI teams collaborate.

Data Lakes

- Low-cost storage
- Supports structured and unstructured data
- Massive scalability

Data Warehouses

- ACID transactions
- High-performance SQL
- Data governance

Core Technologies

- Delta Lake
- Apache Spark
- Photon
- Unity Catalog

DELTA LAKE – THE CORE TECHNOLOGY BEHIND LTAP

Traditional data lakes lack transactional guarantees.

Delta Lake introduces:

1.ACID Transactions

Ensures every operation is completed successfully or rolled back completely.

2.Schema Enforcement

Prevents bad or unexpected data from entering tables.

3.Time Travel

Query historical versions of data.

4.Concurrent Read and Write Support

Multiple users and applications can access data simultaneously without corruption.

This reliability makes LTAP possible.

HOW LTAP ENABLES OPERATIONAL WORKLOADS

Databricks is no longer just an analytics platform. Organizations can now build Operational Applications

- Customer dashboards
- Order tracking systems
- Financial applications
- Inventory management systems

Requirements for Operational Workloads

- Low latency
- High concurrency
- Fast updates
- Reliable transactions

Delta tables combined with Databricks SQL and Photon provide these capabilities.

Applications can directly interact with lakehouse data instead of relying on separate databases.

REAL-TIME DATA PROCESSING WITH LTAP

Streaming + Batch = One Architecture

Traditionally:

Batch Systems → Separate

Streaming Systems → Separate

Databricks unifies both using Structured Streaming.

Benefits:

- ✓ Process millions of events per second
- ✓ Near real-time analytics
- ✓ Continuous data ingestion
- ✓ Unified development experience

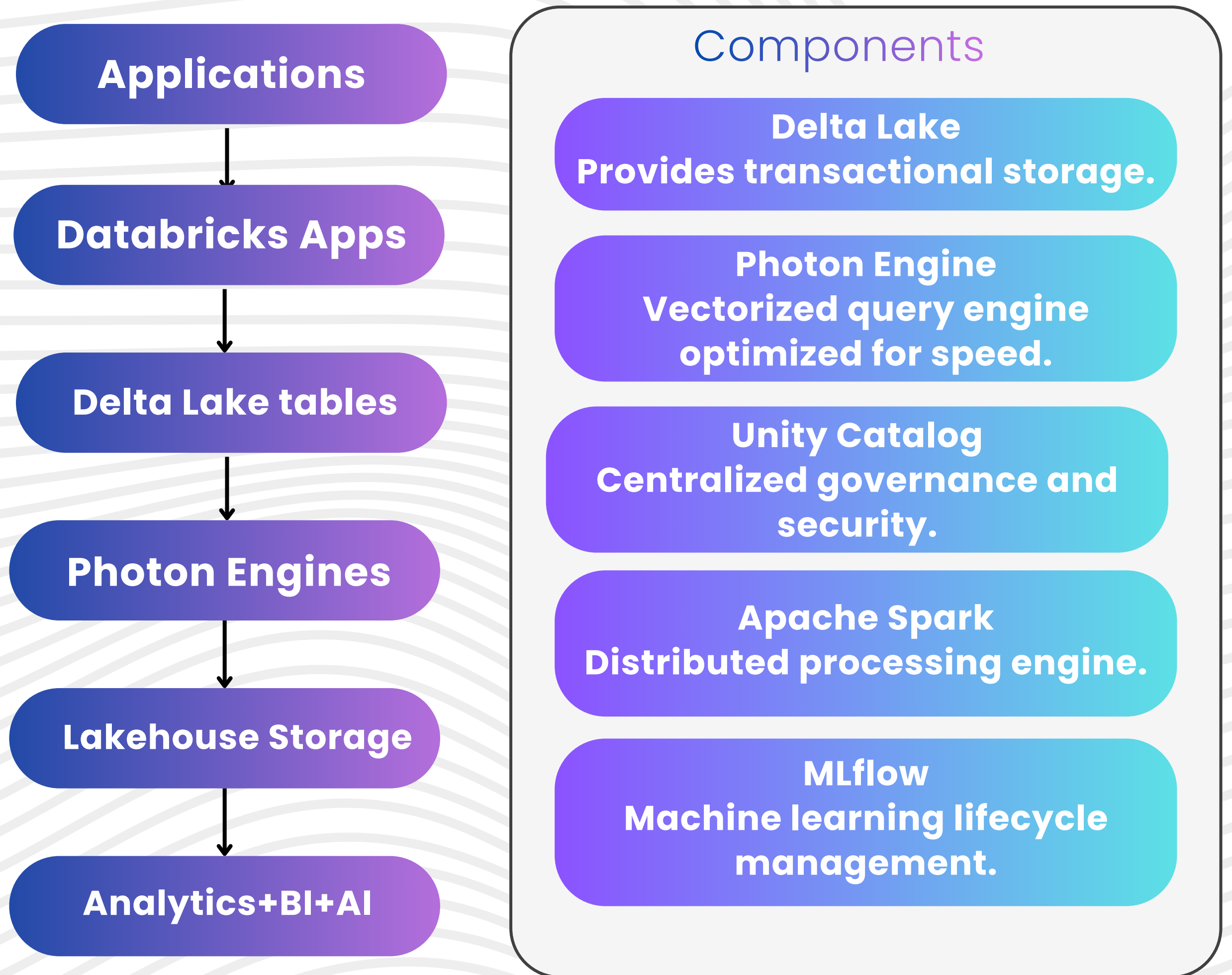
Common Sources:

- Kafka
- Event Hubs
- IoT Devices
- Application Logs
- CDC Streams

Use Cases:

- Fraud Detection
- Recommendation Engines
- Real-Time Monitoring
- Predictive Maintenance

LTAP ARCHITECTURE IN DATABRICKS



BENEFITS OF LTAP

Faster Insights

Analytics can run immediately after data is generated.

Lower Costs

No need for separate OLTP and OLAP systems.

Reduced Complexity

Fewer tools and integrations.

AI Readiness

Data is immediately available for model training and inference.

Scalability

Supports petabytes of data and thousands of concurrent users.

REAL-WORLD LTAP USE CASES

Banking

- Fraud detection
- Real-time transaction monitoring
- Risk analytics

Retail

- Personalized recommendations
- Dynamic pricing
- Inventory optimization

Telecommunications

- Network health monitoring
- Customer behavior analytics

Healthcare

- Clinical analytics
- Predictive diagnosis
- Resource planning

Logistics

- Fleet monitoring
- Route optimization
- Supply chain analytics

Manufacturing

- Predictive maintenance
- Quality monitoring
- Production optimization

KEY TAKEAWAYS

**LTAP = LAKEHOUSE TRANSACTION
PROCESSING**

**BUILT ON DELTA LAKE AND LAKEHOUSE
ARCHITECTURE**

**COMBINES OPERATIONAL AND ANALYTICAL
WORKLOADS**

SUPPORTS REAL-TIME APPLICATIONS

REDUCES DATA SILOS AND ETL COMPLEXITY

**ACCELERATES AI AND MACHINE LEARNING
INITIATIVES**

The background of the image consists of numerous thin, light gray wavy lines that flow across the frame from left to right, creating a sense of movement and depth. These lines are more densely packed in some areas and more spread out in others, giving the background a textured, organic feel.

THANK YOU